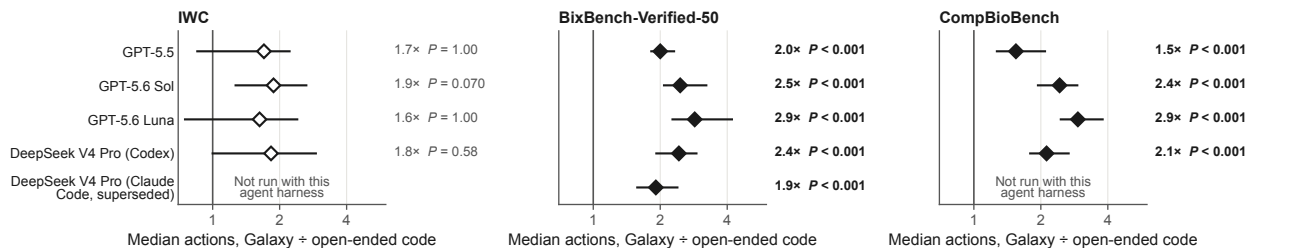
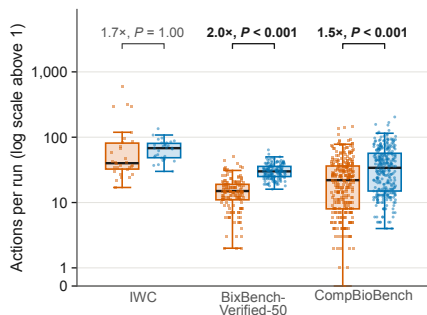


**a** Every model configuration needed more actions in the Galaxy condition on every benchmark; the difference was significant in all 9 comparisons on BixBench-Verified-50 and CompBioBench and in none of the 4 on IWC (ten tasks)



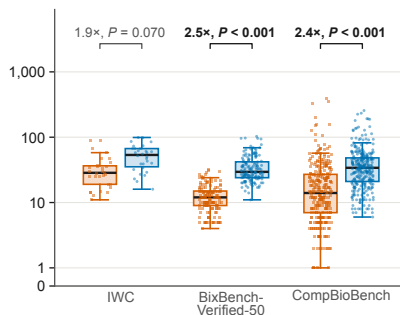
## b GPT-5.5

Galaxy needed more actions on all 3 benchmarks; significant on 2 (BixBench-Verified-50, CompBioBench)



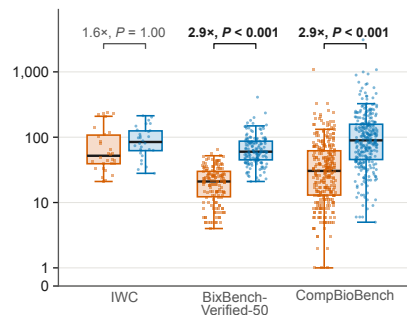
## c GPT-5.6 Sol

Galaxy needed more actions on all 3 benchmarks; significant on 2 (BixBench-Verified-50, CompBioBench)



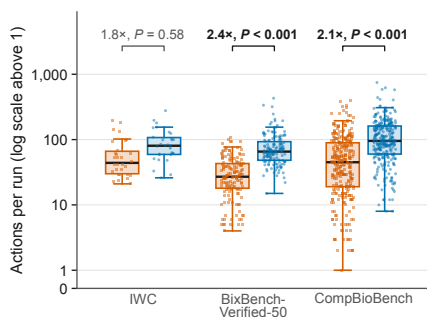
## d GPT-5.6 Luna

Galaxy needed more actions on all 3 benchmarks; significant on 2 (BixBench-Verified-50, CompBioBench)



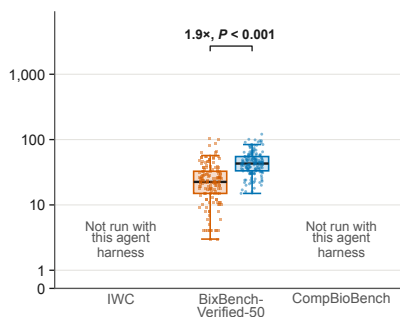
## e DeepSeek V4 Pro (Codex)

Galaxy needed more actions on all 3 benchmarks; significant on 2 (BixBench-Verified-50, CompBioBench)



## f DeepSeek V4 Pro (Claude Code, superseded)

Galaxy needed 1.9 times as many actions (BixBench-Verified-50 only)



Open-ended code condition  
Galaxy condition

Action: one tool call by the agent (shell command, Galaxy interface call, web search or fetch, file read, write or edit); planning-list updates and reasoning steps are not counted. Boxes: middle 50% of runs and median; whiskers: 1.5 times the interquartile range; points: runs (squares open-ended code, circles Galaxy). Ratio: median actions, Galaxy + open-ended code, with a 95% cluster-bootstrap interval (tasks, or source capsules for BixBench-Verified-50; 20,000 resamples).  $P$  value: two-sided Wilcoxon signed-rank test on per-task median actions (each task paired across conditions), Holm-adjusted over the 13 comparisons. IWC: all ten tasks. Excluded: runs without an execution trace (12 CompBioBench), GPT-6 Astra.